

Lecture 4/Module 1

DC Circuit Analysis and Theorems

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Dr. Sudipta Kumar Sarkar

Assistant Professor (Research), SENSE



School of Electronics Engineering (SENSE)
VIT, Vellore, India

Lecture Topics



- Series Resistors and Voltage Division
- Parallel Resistors and Current Division

Series Resistors and Voltage Division

The **equivalent resistance** of any number of resistors connected in series is the sum of the individual resistances.

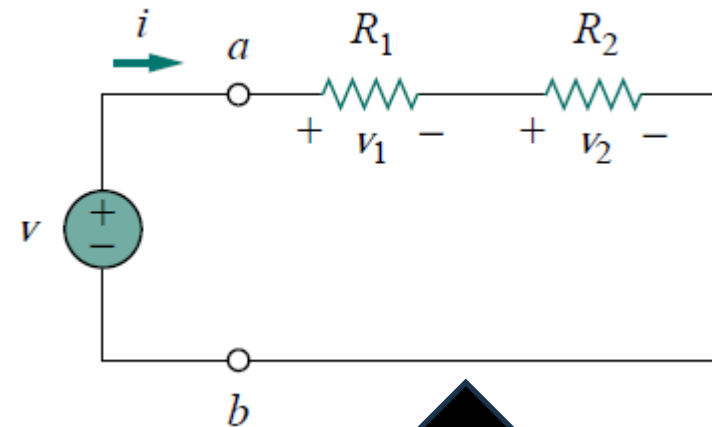
$$v_1 = i R_1, \quad v_2 = i R_2$$

$$v = v_1 + v_2 = i(R_1 + R_2)$$



$$v = i R_{\text{eq}}$$

$$R_{\text{eq}} = R_1 + R_2$$



Voltage Divider Circuit

Voltage Division

$$v_1 = \frac{R_1}{R_1 + R_2} v, \quad v_2 = \frac{R_2}{R_1 + R_2} v$$

Equivalent Resistance & Voltage Division for N nos. of resistors

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

source **voltage** v is divided among the resistors in **direct proportion** to their **resistances**; the larger the resistance, the larger the voltage drop.



Principle of voltage division

Parallel Resistors and Current Division

The equivalent resistance of two parallel resistors is equal to the product of their resistances divided by their sum.

From Ohm's Law

$$v = i_1 R_1 = i_2 R_2$$

$$i_1 = \frac{v}{R_1}, \quad i_2 = \frac{v}{R_2}$$

From KVL

$$i = i_1 + i_2$$



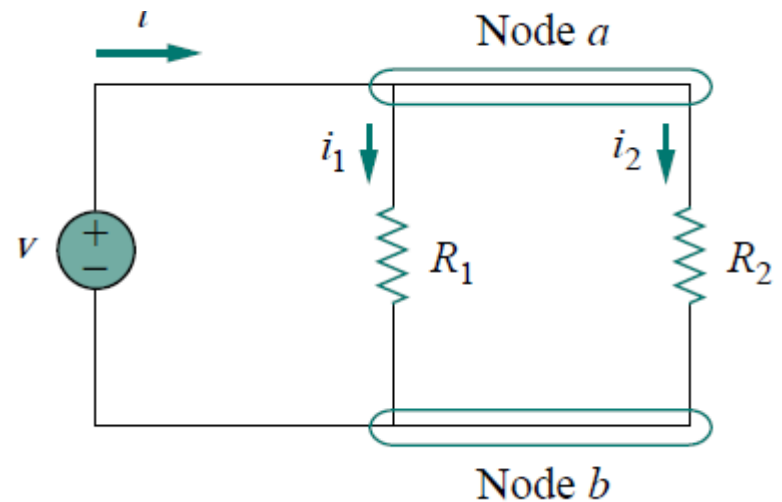
$$i = \frac{v}{R_1} + \frac{v}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{v}{R_{eq}}$$



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

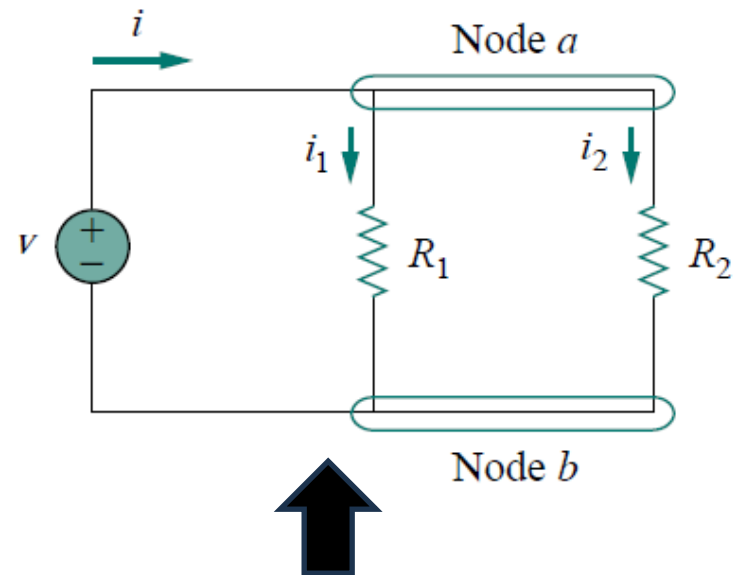
$$\frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



$$i_1 = \frac{R_2 i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

Current Division



Current Divider Circuit

Total current I is shared by the resistors in **inverse proportion** to their **resistances**; the large current flows through the smaller resistance.

What would i_N for N nos. of Resistors connected in Parallel ??

Current Division for N nos. of Resistors

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

Note that R_{eq} is always smaller than the resistance of the smallest in the parallel combination. If $R_1 = R_2 = \cdots = R_N = R$, then

$$R_{eq} = \frac{R}{N}$$

To find out the current division expression for N nos. of Resistors connected parallel it is often convenient to use **conductance rather than resistance**

Equivalent conductance of N nos. of resistors connected in parallel

$$G_{\text{eq}} = G_1 + G_2 + G_3 + \cdots + G_N$$

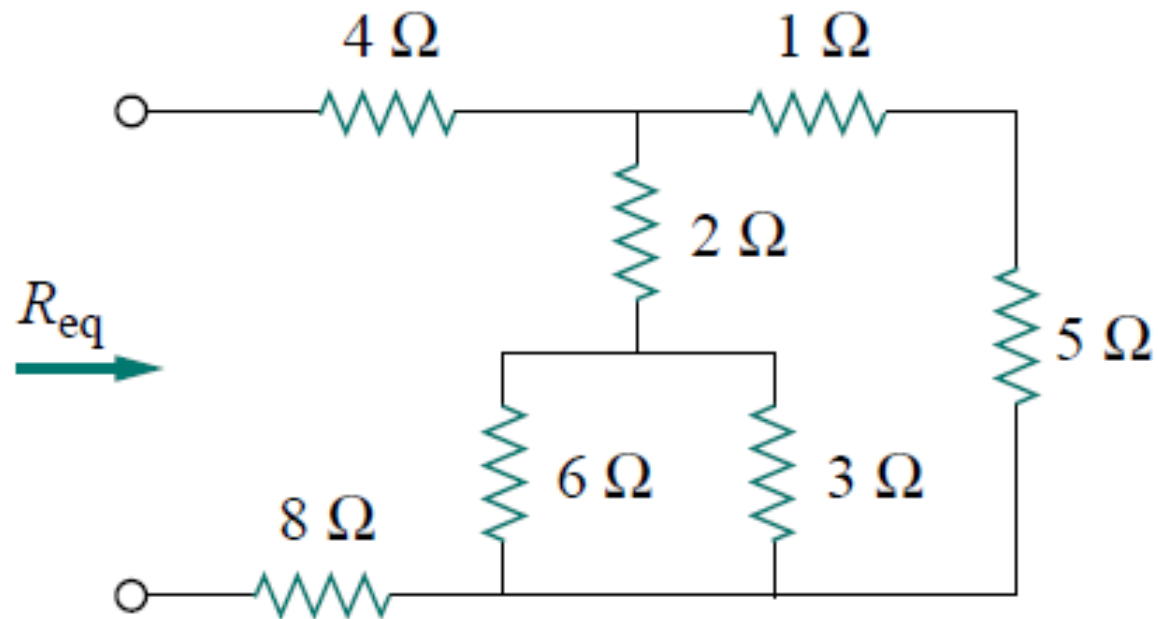
where $G_{\text{eq}} = 1/R_{\text{eq}}$, $G_1 = 1/R_1$, $G_2 = 1/R_2$, $G_3 = 1/R_3$, ..., $G_N = 1/R_N$.

$$i_n = \frac{G_n}{G_1 + G_2 + \cdots + G_N} i$$

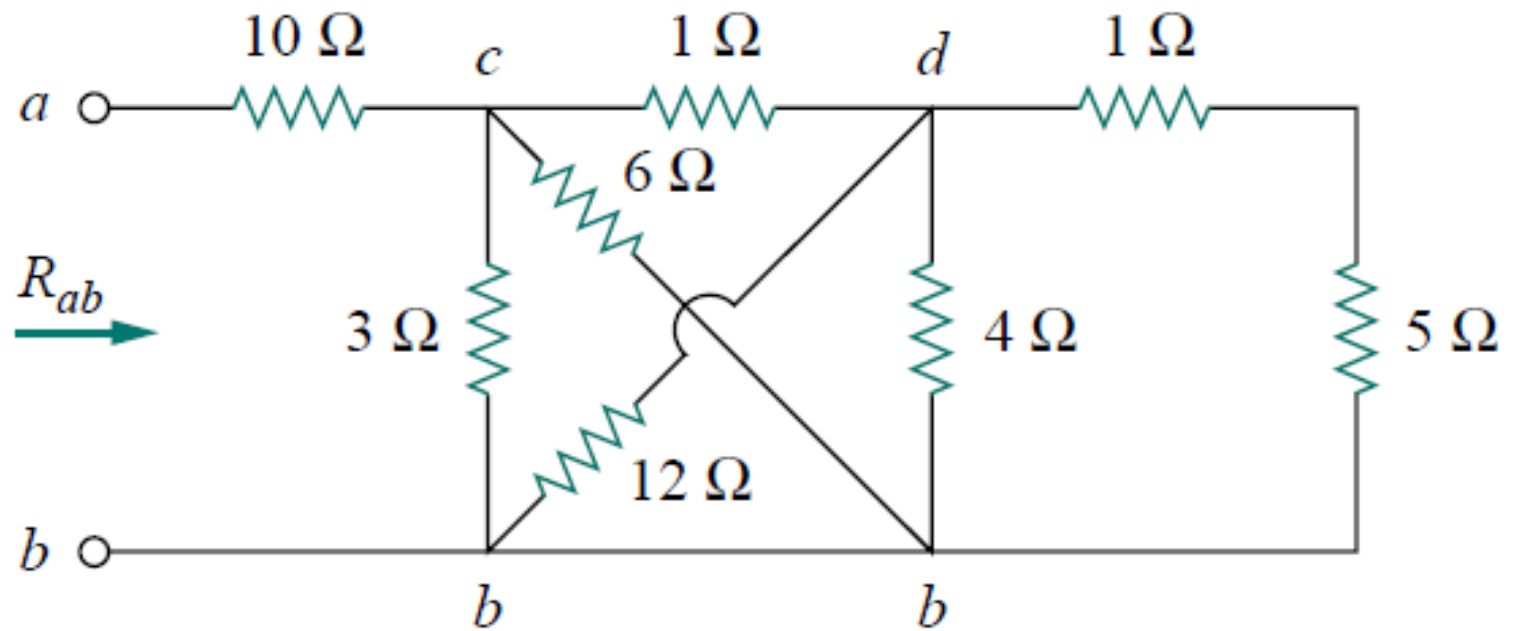


Current Division in terms of conductance

Find R_{eq}

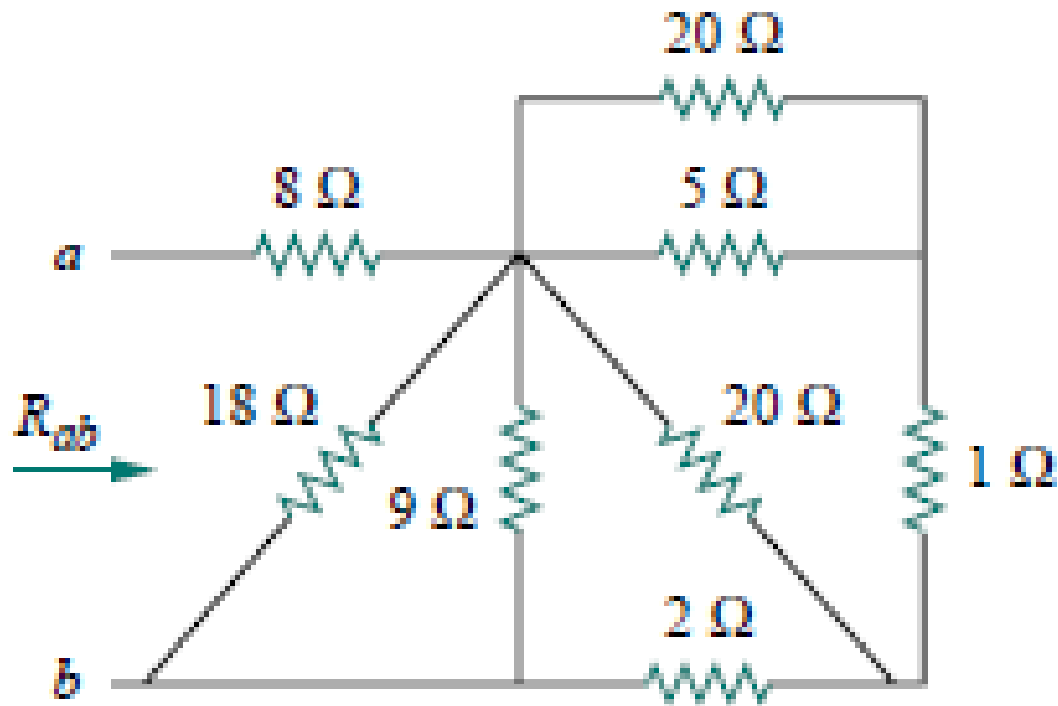


Find R_{ab}

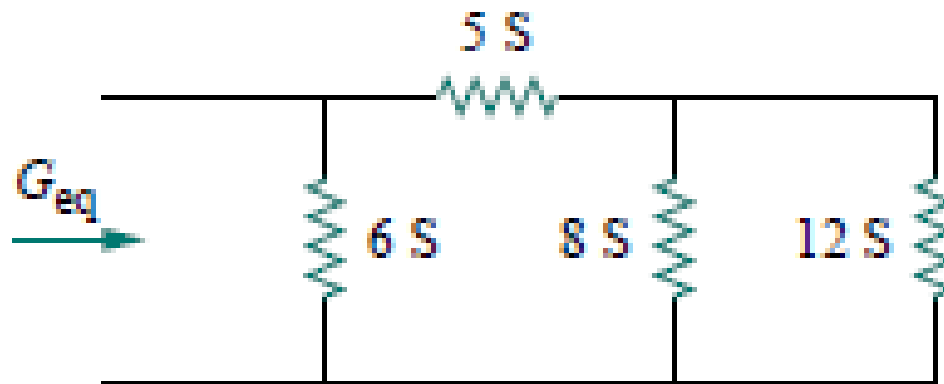


Assignment (Submit by Tuesday 04/07/2026)

Find R_{ab}

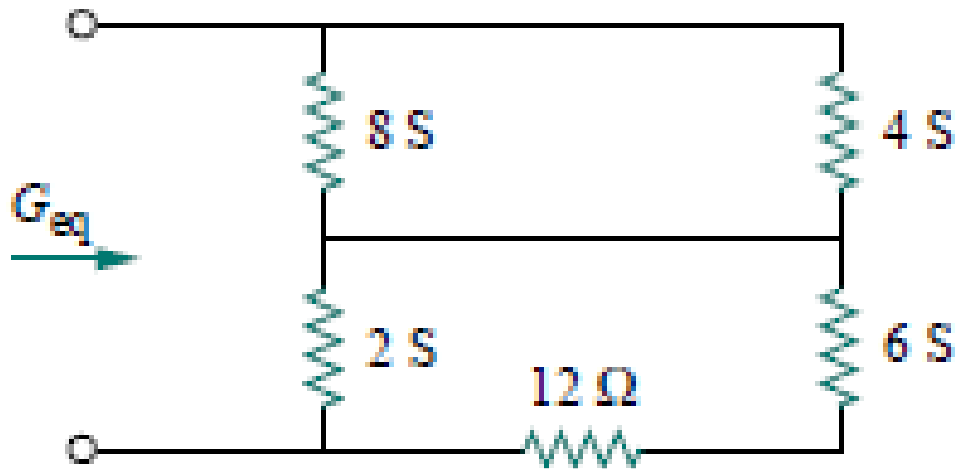


Find G_{eq}



Assignment (Submit by Tuesday 04/07/2026)

Find G_{eq}



Thank You